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Conference Paper · December 2017

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OpenCPU Platform for IoT Applications – A Study

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Abstract— In the past couple of years Internet of Things (IoT) has become a buzzword, even though IoT is not a completely new concept, but a latest iteration of an old concept called M2M (Machine-to-Machine). IoT is in its early stage and is evolving rapidly. A concept which can aid to this evolution is OpenCPU. This paper aims at defining and distinguishing the concept of openCPU, highlighting the features commonly supported by openCPU modules. This work includes description about various application domains where openCPU can be used and also a description about an environment monitoring system implemented with the help of openCPU platform to prove the versatility of openCPU.

Index Terms— OpenCPU, IoT, M2M, remote monitoring, wireless networks, communication.

I. INTRODUCTION

A car engine is able to perform good enough to achieve best fuel economy; this is possible because of the microchip which tells the engine how to perform under various circumstances. Machines are being used to make our lives easier and are becoming smarter. They are smart enough to “talk” with each other, in a language called telemetry. This concept of telemetry is taken to a whole new level with the advent of modern networking technologies. Wireless sensors, internet and computers have come together to achieve machine to machine communication [8]. For such systems, collection of data is carried out using remote sensor units and transmission of data takes place wirelessly through the internet to a server such as a personal computer. At the server side, the data processing followed by the required actions will be performed. M2M communication proves to be a competent technology in many sectors like telematics, energy, security, health, transport and logistics. Irrespective of the sector to which this technology is applied, the core concept remains the same - real time communication between remote machines and centralized data management. M2M has endless possibilities for innovation and also can deliver services with good quality, high speed and at reduced cost. The prominent factors driving the M2M market are growing in range of successful applications, regulatory mandates, and penetration of 3G technologies and advent of 4G technologies. A recent embedded solution that can revolutionize M2M field is *OpenCPU* [1].

II. OPENCPU PLATFORM

A. Conventional and openCPU frame work – A Comparison

OpenCPU is an embedded development solution for M2M field. The OpenCPU platform enables the developer to conveniently design their applications and port the application software to a Micro Controller Unit (MCU), but with a significant change of GSM/GPRS module acting as a main processor. The central idea is to eliminate the usage of a separate MCU. Figure 1 shows the schematic of a conventional M2M frame work where, an external MCU (loaded with the application software) acts an interface between the GSM/GPRS module and the external device. But in an openCPU frame work, the external MCU is eliminated and the application software is ported to the openCPU module as in Figure 2.

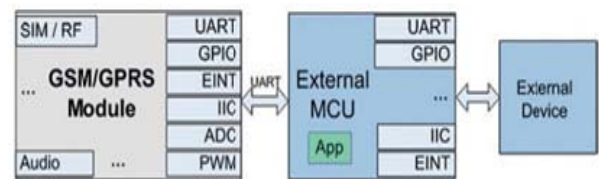


Figure 1. Conventional frame work. Source [2]

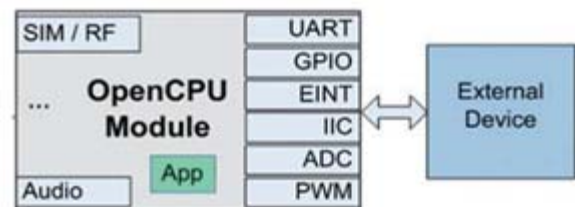


Figure 2. OpenCPU frame work. Source [2]

B. Advantages of openCPU platform

The primary advantage for such an embedded solution is that, it makes hardware design easy and efficient for the developer. The main features of openCPU solutions are:

- Reduction in product development cost
- Reduction in time to market (TTM)
- Simplified circuit design
- Reduced product size and power consumption

C. Supporting features of a typical OpenCPU platform

OpenCPU GSM/GPRS module typically supports:

- ADC and DAC

- Analog audio interface
- SPIs, UARTs and USB connectivity
- I2C bus management
- Serial Display bus
- RTC and Interrupt inputs
- Charging and battery Interface
- PWM / Buzzer interface
- Keypad interface
- Multi SIM card interface
- Programmable clock signal output
- Network & operating status indication
- GPIOs and External memory interface
- Camera interface
- Integrated broadcast FM tuner
- Bluetooth Integrated

The various peripheral support of openCPU platform can be found as in Figure 3.

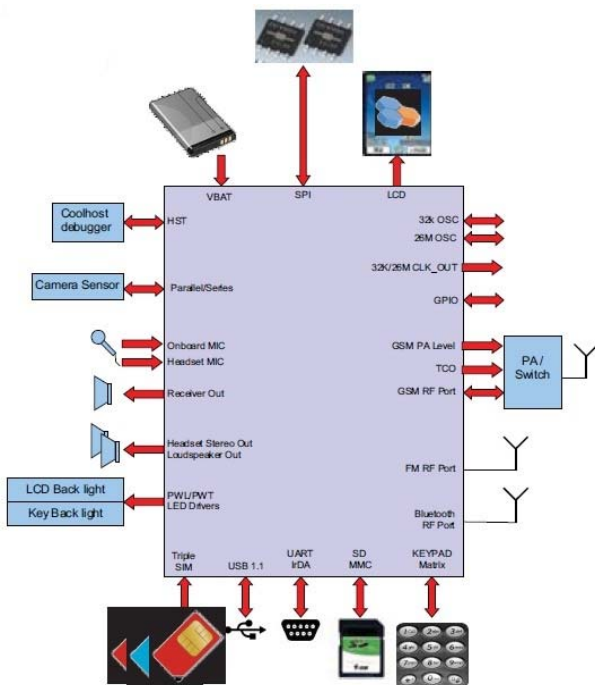


Figure 3. OpenCPU features. Source [3]

OpenCPU software architecture supports the following components through Application Programming Interface (APIs):

- File operations
- Dynamic memory management
- GPIO Pins & Bus control
- ADC / Timer / Local time / System time
- Communication
- TCP / IP, FOTA update interfaces for application
- Standard C Library
- Multitask Management and Inter-task communication

III. STATE OF THE ART

OpenCPU is an emerging platform for embedded applications and it is gaining popularity among academicians as well as among industry personnel. In order to assess and understand the reach of openCPU platform, a study has been done. A project thesis: '*Remote measuring and logging - Evaluation of the Quectel openCPU technology*' aims at improving the existing precision agriculture product from Sensefarm (based in Lund, Sweden), a company which develops embedded solutions for agriculture is studied [4]. The existing product was developed to collect, store and transmit data on local environmental conditions of a farm. This solution makes use of a separate Arduino controller at the base station where information is acquired and is equipped with a separate GSM module for data transmission. To make the product more affordable, the functionalities of the GSM module and the Arduino controller were combined. This integration was brought about with the help of M66 GSM module from Quectel, which supports openCPU and allows programming of a custom firmware which can take over the responsibilities of the Arduino controller. The overall aim of this thesis is to investigate whether M66 GSM module with the support of openCPU is able to replace the existing solution, and it was done by evaluating the following criteria:

- Three years battery life without human intervention
- Power off and wake up at set time
- Reception of updates using Firmware Over The Air
- Transmission and reception of data
- Communication with I2C devices
- Positioning data acquisition from GPS device

Of the above six criteria, five criteria were met with the help of the following features:

- Hardware Interface API
- Socket API along with code implementation for HTTP and JSON
- Built in support for FOTA
- RTC modules

The criterion of battery life was met partially, but in a theoretical sense this constraint was achieved. The thesis concluded on a positive note, stating that openCPU solution could be an effective replacement for the current processor-GSM solution.

In [5], a *GPS based personal tracking system* was designed using the openCPU platform. The primary objective of this work is to make the device accurate, less in weight and compact. This is achieved by the usage of openCPU platform, thereby eliminating the need for a separate processor. OpenCPU based approach reduced the product development period, simplified the circuit design, decreased the product size, decreased the total cost and power consumption.

IV. APPLICATIONS USING OPENCPU PLATFORM

A. Automotive On-Board Diagnostics

On-board diagnostics (OBD) refers to a vehicle's self-diagnostic and reporting capability [6]. It gives the health information for various vehicle subsystems. The OBD-II is a standard which is implemented in all vehicle models developed from 1996. This standard is used to measure the real-time data of various engine parameters like coolant temperature, engine rpm, etc. By sending appropriate request messages to the OBD of the car, these parameters can be measured. The OBD also provide the diagnostic trouble codes (DTCs). They are stored in the engine control module (ECM) memory and indicate the circuit in which fault has occurred. When the request message for DTC is sent, the ECU responds with the available DTCs stored and thus the vehicle health status can be observed.

On-board diagnostics can be sent to backend server using CAN transceiver, CAN controller and Open CPU-GSM Modem. Open CPU GSM Modem operates as a central controlling device; it controls the ECU access through serial interface with CAN controller and transfers log to back-end server. OBD solution using openCPU allows manufacturers to create quick end products.

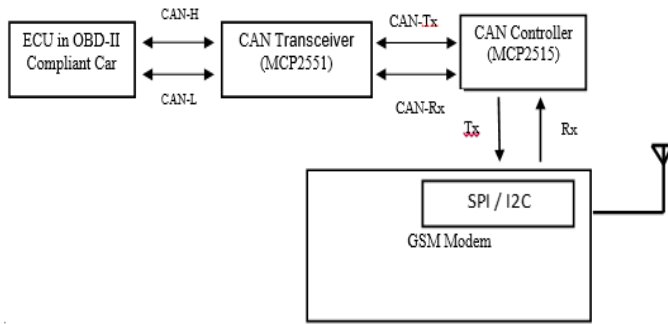


Figure 4. OBD solution using openCPU

B. Temperature/ Humidity Monitoring Station

Measurement of temperature and humidity play a crucial role in places where the environment has to be controlled, for example, in places like milk dairies or building management system (BMS). In case of milk dairies, temperature and humidity conditions play an important role in storing and pasteurizing milk. OpenCPU GSM modem can be used to extract data from temperature and humidity sensor through channel ADC (10-bit resolution). This data can be transmitted to the diagnostics for analysis, using which the temperature and humidity can be controlled for best results.

In case of BMS, depending on the temperature and humidity values read from the sensors, openCPU GSM modem can be programmed to automatically control air conditioning unit so as to maintain specific temperature and humidity conditions.

C. Remote monitoring and controlling of equipment

Low cost monitoring system can be used for Potential Free Contact (PFC) and non-PFC alarm detection and controlling of site equipment (such as air conditioner, exhaust fan, battery, generator etc.) by driving relays through GPIO. Industrial meters can be connected through RS485 to the alarms to collect data on regular intervals. Several monitoring devices exist in the market. However, not only are they application-specific, but also are expensive. Hence, there is a need for a low cost system that can be implemented in large numbers for various applications for effective monitoring. Additional diverse applications which demand the support of remote monitoring and control include [7]:

- Smart metering: Green energy solutions
- Vehicle tracking with GPS and without GPS, done through triangularisation method on the multiple tower co-ordinates
- Asset monitoring
- Mobile Health (m-Health)

D. Internet of Things (IoT)

IoT is an emerging technology, an extremely broad field influencing various sectors viz., infrastructure, logistics, retail, transpiration etc [9]. Data collection is done with the help of these devices and communication takes place autonomously between other devices. Such useful data can be collected on an openCPU device through network and can be analyzed. This data can be used for improving the quality also to study pattern and predict probabilities of various events. For all of the above mentioned applications, a server can be used to

- Communicate with thousands of clients
- Track & control the status of clients
- Upgrade firmware, thus reducing OPEX (Operational Expenditure)

The communication can be facilitated using TCP or UDP or HTTP or HTTPS.

V. AN IoT APPLICATION DEVELOPMENT USING OPENCPU PLATFORM

A Server User Interface was developed for 'Environment Monitoring System' implemented using openCPU platform. A system of six clients (mix of Quectel & Fibocom openCPU modules) were configured to measure temperature and humidity of the environment and send the data to server along with co-ordinates (latitude and longitude) of the place where they are stationed. Using these co-ordinates the clients' position was mapped through Google map. For firmware up-gradation, most of these modules have built in features (normally use TR69 specification), thus eliminating the software components at server as well as client side.

The Figure 5 shows a user interface that can be used for observing the position of various monitoring units, status and so on. The above mentioned low cost monitoring systems are versatile hardware that can be used for monitoring and control in various environments and for various applications. They are cost effective and can be deployed in large numbers.

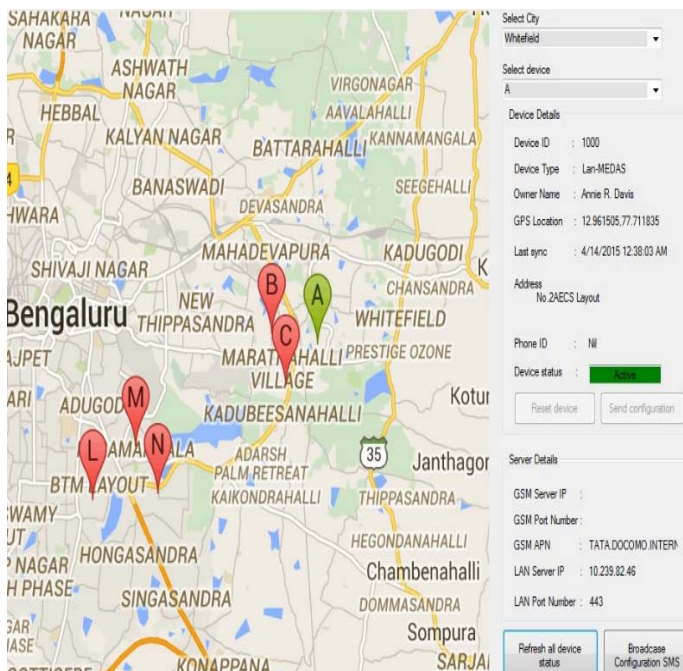


Figure 5. User interface for temperature and humidity monitoring system using OpenCPU platform

Remote controlling and configuration is handled through GPRS and SMS. Live interaction with these clients and configuration update done from Server through GPRS connectivity. Scaled down configurations and status are exchanged through SMS from a set of registered/authenticated mobile numbers. By this SMS commands, the user can know the status of the devices without server assistance. Normal configurations include like, changing IP address, port number, threshold level changing for alarms. The clients used to interact with server on two modes: 1. Periodic mode and 2. Event based mode. In the periodic mode, the clients used to send the GPRS data to server on a configured intervals time. In case of event based mode, the GPRS data sent on occurrence of an event (like status change or alarm or critical events). Firmware upgradation gets triggered through SMS message indicating the server IP & port number to be used for new software and then GPRS connection will be established. In case of network fail, the client stores the data locally and will be delivered to server in a chunk once the network is up. Each message will carry message creation time stamp and message ID.

VI. OPENCPU ARCHITECTURAL FEATURES INFLUENCED BY THE APPLICATION DOMAIN

- These Open CPU modules support boot time configurable option for either Open AT commands or directly use APIs from the abstraction layer.
- They have RTOS as well; the development environment provides complete tool chain with RTOS.
- Instead of controller and 2G/3G modules as two separate components, it exposes the GSM module's shared memory and CPU space for a user to accommodate his logic or solution. This helps battery operated devices for efficient power usage.
- Possibility of remote debug and software upgrade makes it cost effective and thereby reducing the OPEX.
- Reduced energy footprint of the variable and battery operated devices.

VII. CONCLUSION

The concept of openCPU is bestowed with the potential to improve various M2M and IoT applications by reducing the cost, the product development cycle and the form factor. Most of these modules come in the form of MIB (Module In Box) and are ready to use. Knowing the architectural and peripheral support of openCPU platform, one can develop IoT and M2M products and applications with reduced cost and avoiding the burden of interfacing between MCU and the GSM module. This can result in a cost effective product development cycle for various industrial applications. The versatility of openCPU platform is demonstrated through the IoT application – '*environment monitoring system*' using six clients (openCPU modules) and a single server through the *server user interface*. The obtained results are in par with the conventional MCU-GSM frame work and outperformed the conventional system in terms of reduced cost and reduced interfacing complexities. This guarantees that when openCPU technology is deployed at a larger scale for big and complex data streaming and IoT applications, will maximize the profit and minimize the product development time. Thus, it can be foreseen that the concept of openCPU will revolutionize the fields of M2M and IoT.

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